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To SFC's Licensed Corporations

Advice to Investors on Exotic Warrants

The SFC issued a press release on 26 June 2003 to give investors general advice on exotic warrants. We write now to remind licensed corporations that, pursuant to the requirements under the Code of Conduct for Persons Licensed by or Registered with the Commission, licensed corporations dealing with clients who trade in exotic warrants should ensure these products are suitable for your clients and assure yourselves that your clients understand the nature and risks of these products. For more details, you should refer to the Code of Conduct which is available on the SFC's website www.hksfc.org.hk.

The SFC has recently received a number of complaints over the performance of exotic warrants and most of these complaints seem to have arisen from a lack of understanding of the product features, and the risk and return profile of exotic warrants. For instance, some investors were not aware that a "capped call warrant" imposes a limit on the upside profit potential and complained that the warrant price remained stagnant even though the underlying stock price had moved up. Other investors confused a "locked-in return warrant" with a standard warrant and alleged that it underperformed the standard warrant when the price of the underlying stock surged. It turns out that the locked-in return warrant was out of the money at past fixing dates and, therefore, it had already lost some of its value.

We have also noticed that some investors do not fully understand the role of liquidity providers. Although issuers bear the responsibility of providing liquidity in their warrants via liquidity providers, this does not imply that they are obliged to take up all the outstanding orders in the market. Issuers provide quotes via liquidity providers at the price level they deem fair and within the maximum spread to which they have committed in the respective listing document. Therefore, there is a chance that their quotes may not match investors' orders or expected price levels.

In the above press release, investors were reminded to read the listing documents for exotic warrants and understand the terms and conditions. This is to ensure that they are aware of each issue's unique features and understand how it works before making an investment decision. Investors were also advised to seek professional financial advice if they have any doubt.

To help investors learn more about exotic warrants, we have published an educational article on *Exotic Warrants*. To facilitate your explanation of the characteristics of exotic warrants to



your clients, we attach below that information for your reference. We have also published other educational materials on derivative warrants and the role of the liquidity providers on the “Features” section of the SFC-operated Electronic Investor Resources Centre at www.hkeirc.org.

Investor Education and Communications Department

Investor Education

Exotic Warrants

When you are checking on warrant prices on the Teletext, have you ever noticed that some derivative warrants carry a symbol “X” in their names? Do you know what that symbol means?

The symbol “X” refers to a special class of derivative warrants called “exotic warrants”. As the name implies, exotic warrants are derivative warrants with exotic features that are different from standard call or put warrants. Some of them are modified single call or put warrants, while others are effectively a combination of two or more standard warrants.

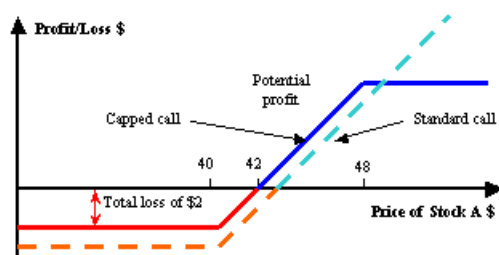
How do exotic warrants work? What should I be aware of?

Make sure you know the salient features of an exotic warrant and understand how it works:

Each type of exotic warrant carries unique features and behaves quite differently from standard warrants and other exotic warrants in its response to the price levels or price movements of the underlying asset. Here, we give some examples.

Capped warrants: These limit or “cap” your potential profit but are generally cheaper than standard warrants which give you unlimited upside profit potential. They can be call or put warrants and are sometimes known as call spread warrants and put spread warrants. Suppose you buy a capped call warrant at \$2, on underlying stock A with an exercise price of \$40 and a cap level of \$48. If the price of stock A is above \$42 (the breakeven point) on the expiry date, you can make a potential profit as illustrated in Diagram A. Generally, the higher the stock price, the higher your return. However, unlike a standard call warrant which has unlimited upside profit potential, your maximum potential profit is “capped” at \$6 even if the stock price rises beyond \$48.

Diagram A. Capped call warrant vs Standard call warrant



An investor who buys the above capped call warrant is effectively buying a call warrant at an exercise price of \$40 and selling a call warrant at a higher exercise price of \$48. The premium from the warrant “sold” makes the capped call warrant cheaper than a standard call warrant.

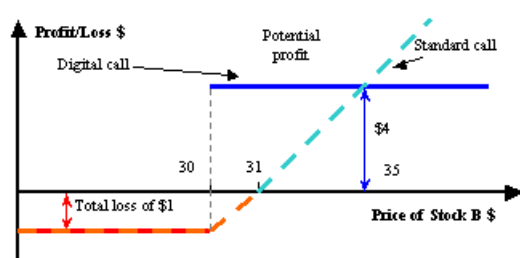
Locked-in return warrants: These are usually warrants with longer term (e.g. 18 months) maturity at launch. They take snapshots on regular (e.g. quarterly) fixing dates and will “lock-in” the profits if the warrants are in the money on those fixing dates. The average of



profits locked-in on different fixing dates are then paid to you at expiry. This is to allow you to participate in the upside of the warrants during their lives even though they may expire out of the money. Locked-in return warrants are basically a combination of a series of warrants which expire on their respective fixing dates but their settlements are delayed until the final expiry date.

Digital warrants: Payoff at expiry for standard warrants varies at different levels of the underlying price. On the other hand, digital warrants pay you either a “fixed” amount of cash or nothing, depending only on whether the warrant is in the money at expiry. Suppose you buy a digital call warrant at \$1, on underlying stock B with an exercise price of \$30 and a potential payout of \$5. If the price of stock B is at or above \$30 on the expiry date, you can make a “fixed” potential profit of \$4 as illustrated in Diagram B. If the stock price is below \$30, you will lose all your initial outlay. Also, if the stock price is at \$50, you still only make \$4, as the warrant price will not rise as much as a non-digital warrant as the stock price rises.

Diagram B. Digital call warrant vs Standard call warrant



Dividend accumulator warrants: These are usually warrants with longer term (e.g. 18 months) maturity at launch. They pay you dividends similar to the ordinary cash dividends paid on the underlying share during the lives of the warrants prior to their expiry dates.

To know more about the features of an exotic warrant, get a copy of its listing document from the [HKEx website](http://www.hkex.com.hk). It sets out all the terms and conditions of the warrant. Some warrant issuers also provide listing documents, term sheets or other marketing materials on their own websites for public access.

Understand the pros and cons of an exotic warrant to see whether it meets your investment needs: Dealing in warrants involves risks that may be different from investing in the underlying securities. Like standard warrants, exotic warrants offer a leveraged exposure to movements in their underlying assets. Some exotic warrants offer more leverage than standard warrants while others offer less. Remember, leverage works both ways. Although higher leverage may give you a higher return, it also exposes you to a higher risk.

Besides, each type of exotic warrant has its own advantages and disadvantages. For example, a capped call warrant is normally cheaper than a standard call warrant with the same exercise price and expiry date. This is because you forgo any profit potential above the cap level (see Diagram A). While a dividend accumulator warrant lets you enjoy dividends paid on the underlying share, it certainly costs more than a standard warrant. A digital call warrant will outperform a standard warrant if the underlying price is at or within a certain range above the



exercise price at expiry, but will underperform if the underlying price goes up further (see Diagram B).

Ensure you know the differences between an exotic warrant and a standard warrant:

Because of the difference in its payoff structure, the change in the warrant price of an exotic warrant in response to movements of the underlying asset is also different from that of a standard warrant. Make sure you fully understand the risk and return profile of an exotic warrant before making any investment decisions. Do not expect it to behave in the same way as a similar standard warrant. If in doubt, seek professional financial advice.

Be aware of different symbols used by different newspapers for identifying exotic warrants:

Some newspapers use the symbol “X” to identify all types of exotic warrants in their pricing page. Others adopt the symbol “X” to represent only one type of exotic warrant, say dividend accumulators, and use other symbols to differentiate other exotic warrants. So, when you are checking warrant information in newspapers, read the legend first and understand what the symbols represent. Do not mix up different types of exotic warrants.

Are commonly used analytical tools like premium and gearing applicable to exotic warrants?

They should be used and interpreted with care. Exotic warrants are sometimes a combination of several warrants. A simple calculation using such analytical tools may sometimes give misleading results. For example, the premium of a standard warrant is defined as the percentage by which the warrant price exceeds its intrinsic value (i.e. spot price minus exercise price). It can also be viewed as the time value of a warrant. If we use the same formula to compute the premium of a capped call warrant, an anomaly of “negative” premium may sometimes occur. Why?

This is because a capped call warrant effectively consists of buying a call warrant at the lower exercise price (long call at the strike level) and selling another call warrant at the higher exercise price (short call at the cap level). The value of the capped call warrant is the value of the long call warrant minus the value of the short call warrant with each warrant being valued at the sum of the intrinsic value and the time value. Therefore, the premium or the time value of the capped warrant (T_c) is equal to the time value of the long call warrant minus the time value of the short call warrant ($T_1 - T_2$). In the event that the time value of the long call warrant is lower than that of the short call warrant, this may result in “negative” premium.

Should I buy exotic warrants?

As in making other investments, you should consider carefully whether a particular exotic warrant is suitable for you in light of your experience, investment objectives and financial position.

Exotic warrants are designed for those who have a clear view and strategy on the movement and level of the underlying asset. For instance, a capped call warrant is a cheaper tool for



investors who believe that the underlying price will only move up by a certain range. Dividend accumulator warrants are more investment style instruments. They target investors who wish to have a longer term exposure to movements in the underlying stock and to receive dividends but are happy to pay for the potential benefits.

After all, exotic warrants are for experienced investors who are willing to spend the time to understand the nature of the products so that they can choose the one which meet their market view and investment objective. Do not buy a warrant simply because it looks cheaper than the others, without considering why it is cheaper and whether it suits your needs.

